

REPORT – E-COMMERCE RFM & CUSTOMER BEHAVIOR ANALYTICS

This report delivers an exhaustive analysis of consumer shopping habits and transactional performance for an e-commerce platform. It covers exploratory data analysis, relational SQL querying, an RFM segmentation model, and an interactive Power BI dashboard built from 120,000 transactions across 5,000 unique customers (full year 2025).

E-E-COMMERCE-QY
ANALYTICS NO:5



Project Overview

Business Objective

A leading retail company aims to understand customer shopping behavior to enhance sales performance, customer satisfaction, and long-term loyalty.

Key Challenges

Identifying changing purchasing patterns across demographics, product categories, and sales channels to inform targeted strategies.

Core Question

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"



InvoiceNo	CustomerId	InvoiceDate	UnitPrice	TotalAmount
#4296CD	CustomerId	Quantity	UnitPrice	#45190
	QuanimeDatt	#2589C9	249.00	#3890
	#1128PGF	600	12,19%	21956
	CustomerId LVce ArvoceDs	e5f6ff0	18,190	2,60%
	Quaisl Cvaolce	#e5190%	16,60%	35610
	Coainalerlb	#557640	#9,0%	#39%
	Qustoniert LVce Quantrice	000	#5,00%	35.90
	#0atorfclld	#339600	45,16%	24#55
	1E93EFB	#551000	18,10%	2E3658
	Cualcenef Country	200	45,190%	#5.8%
	10afoclf QuaicnBis	600	45,9%	216610
#22C3B	Conicle5HF	#251099	5700%	#86656
#063E5F	#1EFBBFF	#1631000	1550%	248%
	T#EE3C4EF		5,18%	#34830
	TotalAmount	#15,080	3,16%	38675
	#2E3C4FE TivalceDs	#25,900	38110%	250%

Dataset Summary

The dataset comprises 120,000 transaction records from 2025-01-01 to 2025-12-31 across 5,000 unique customers. Key schema fields: InvoiceNo, CustomerID, InvoiceDate, Quantity, UnitPrice, Country, and TotalAmount. Geographic distribution is dominated by the United Kingdom (~75%), with Germany, France, EIRE, Spain, and the Netherlands also present.



Exploratory Data Analysis (Python)

We loaded and cleaned data with pandas, handled missing values, and converted types. RFM features were engineered by grouping on CustomerID: Recency (days since last purchase), Frequency (transaction count), and Monetary (total spend). Customers were segmented into cohorts: Champions, Loyal Customers, At Risk, Hibernating/Lost, Recent/New, and Promising. Cleaned datasets and RFM metrics were exported to PostgreSQL (fact_transactions, dim_customer_rfm, agg_rfm_summary) for SQL analysis.

RFM Summary Statistics

Summary of RFM metrics across 5,000 customers (counts, central tendency, spread): mean Recency ~12500.5, mean Frequency ~15.8, mean Monetary ~13738.91. Distribution ranges: Recency min 10001 to max 15000; Frequency 1–141; Monetary 3125.07–32080.5. These statistics underpinned score bucketing and cohort assignment.



Recency

Mean 12500.5 days; range 10001–15000



Frequency

Mean 15.8 transactions; range 1–141



Monetary

Mean 13738.91; range 3125.07–32080.5

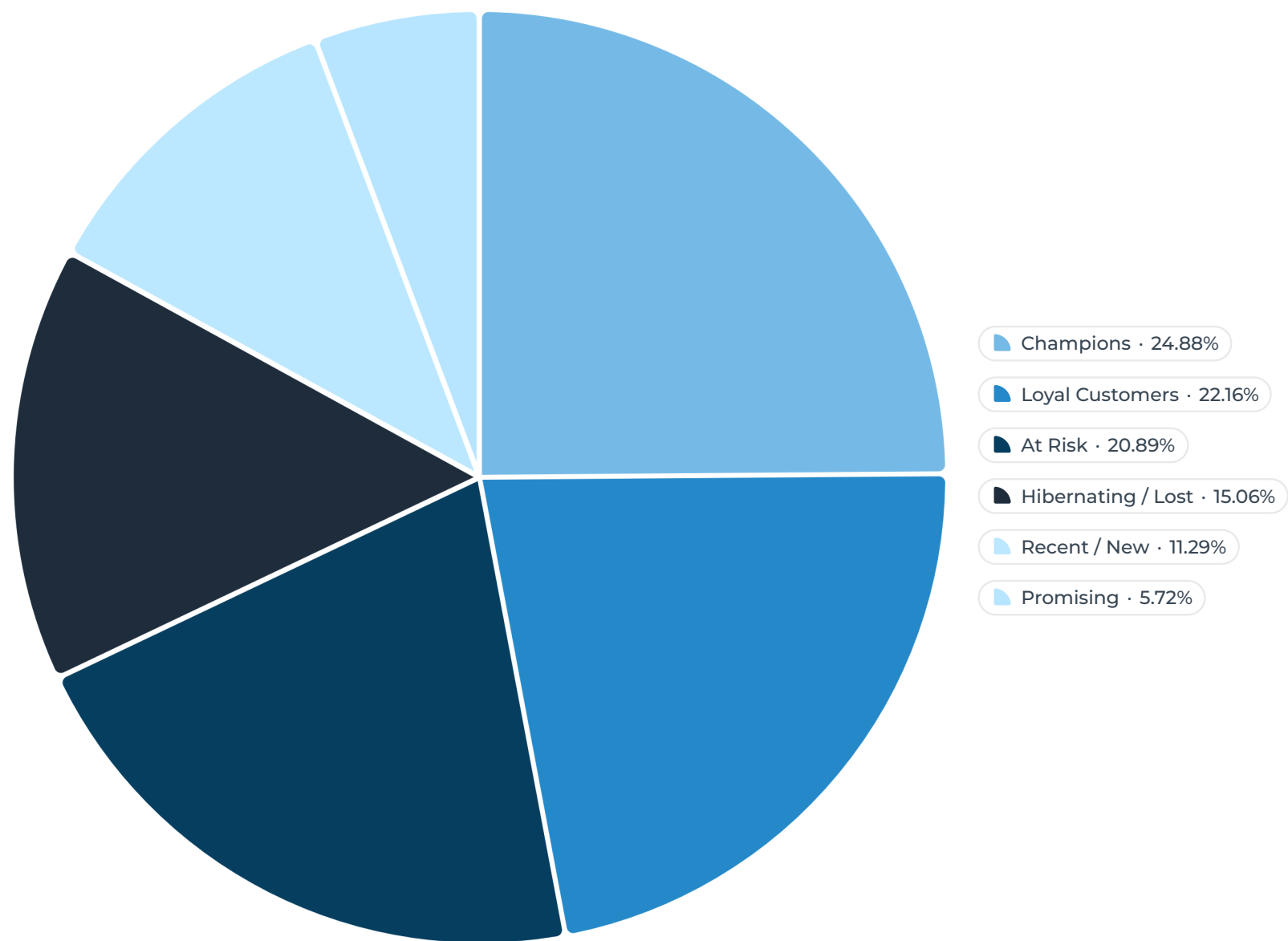


RFM Profiles & Scoring

R, F, M scores were calculated per customer and concatenated to form RFM_Score and RFM_Total. Example profiles (first five customers) show varied Recency, Frequency, Monetary values and resulting RFM scores. The RFM scoring completed successfully and enabled cohort mapping for targeted actions.

Segment Distribution & Revenue Share

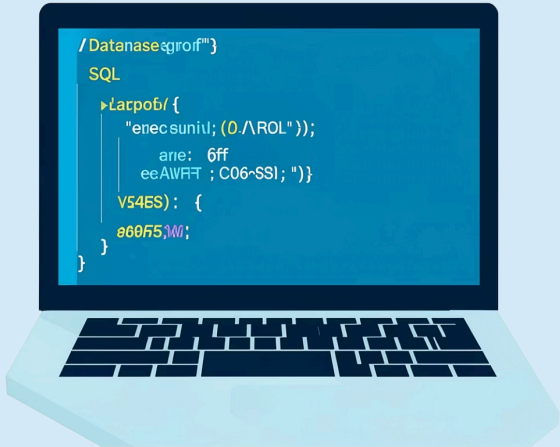
Segment distribution and revenue contributions reveal concentration in high-value tiers. Highlights: Champions (1,028 customers) contribute ~24.88% revenue; Loyal Customers (1,036) contribute ~22.16%. Other segments: At Risk (936), Hibernating/Lost (949), Recent/New (699), Promising (352). These cohorts drive strategic prioritization for retention and upsell.



The pie chart shows relative revenue share by RFM segment to emphasize where monetization is concentrated.

SQL Analysis – Key Business Questions

We executed PostgreSQL queries to answer operational questions: revenue contribution by cohort (Champions 24.88%, Loyal 22.16%), relational integrity checks across fact and dimension tables, recency & frequency distributions by region, and isolation of At-Risk customers using high recency and low monetary filters for churn mitigation.



Result:	#2589CD			Result:	SQD		
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#0158CCB	30	16)		#09	7	80	\$0 18.1
1E23D	80	90		21	10	9	\$0 0610
063E5F	50	32		21	53	18	\$0 0310
F4FBSB F1F6l	36	38		22	33	90	\$3 0640
E27638 CQL	80	37)		14	40	10	\$6 0610
2E3C4E S11SV;	513	(2)		13	36	40	33 2710
2E3C4F 88/553	00	(5)		23	30	10	10 0617

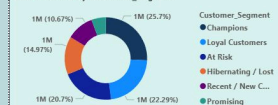
E-Commerce Customer Behavior & RFM Analytics

Total Customers
5K

Total Revenue
6M

Total Orders
8K

Total Revenue by Customer_Segment

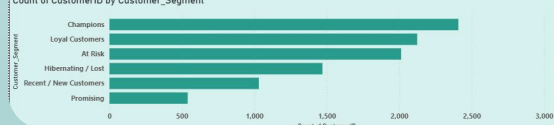


Customer_Segment
● Champions
● Loyal Customers
● At Risk
● Hibernating / Lost
● Recent / New C...
● Promising

Sum of TotalAmount by Month



Count of CustomerID by Customer_Segment



Country

Italy

France

Germany

Netherlands

Spain

United Kingdom

Power BI Dashboard & KPIs

An interactive Power BI dashboard centralizes KPIs and data models. Key metrics displayed: Total Customers = 5K, Total Revenue $\approx$ 6M, Total Orders $\approx$ 8K. Visuals include a donut chart for revenue by customer segment, monthly revenue trend (consistent $\sim$ 0.5M per month), and horizontal bars for customer counts by segment. Filters allow dynamic slicing by country, date range, and RFM segment for executive reporting.

Conclusions & Recommendations



Protect High-Value Segments

Reward Champions and Loyal Customers with VIP perks, early access, and personalized recommendations to maximize LTV.



Re-engage At-Risk Accounts

Launch automated win-back campaigns with time-sensitive discounts for At Risk and Hibernating/Lost segments to recover churned revenue.



Optimize Retention Workflows

Implement lifecycle triggers based on Recency to reach customers before they slip from Loyal to At Risk.



Localized Regional Campaigns

Capitalize on strongholds (United Kingdom 75%) with localized promotions and shipping incentives for secondary markets like Germany and France.

- ❏ RFM analysis shows revenue concentration in high-value tiers and clear churn vulnerability—prioritize retention and targeted engagement.